

Sylvia Mirecki

Software Engineer

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Work Experience

- Sep 2023–Jun 2026 **Co-founder / CTO**, *Searchless*, Toronto, Canada.
Full Time
- Searchless was an AI powered video interviewing company I co-founded. During my tenure, I worked with dozens of clients, including going on a trade mission to Thailand with the Trade Commission of Canada. Unfortunately, we didn't hit profitability after 3 years.
 - Implemented real time multimedia (video/audio) processing code in Rust with a stream based recursive descent parser (mostly Rust but a bit of TypeScript for the frontend component).
 - Wrote the response generation logic for interviewing job candidates (Rust, Neo4j, AWS, Postgres, LLM APIs with a bunch of fun RAG stuff)
 - Hired and managed 3 engineers
 - Aligned our software to PDPA (and somewhat GDPR + SOC2) compliance
- Aug 2022–Sep 2023 **Software Engineer**, *Dropbox*, Toronto, Canada.
Full Time
- Wrote technical feature proposal docs
 - Worked across 3 Go services and 2 Rust services
 - Solving distributed system failures through metrics and massive log stores
- Jan 2020–Jul 2022 **Software Engineer**, *Amazon Alexa*, Arlington, USA / Toronto, Canada.
Full Time
- Wrote design docs for RDF graph schemas, collaborating with linguists
 - Redesigned ETL pipeline to cut back 3p onboarding time from weeks to days via Scala code generation
 - Redesigned caching mechanism with Redis to reduce costs 5x-20x
 - Hosted tech talks on parsing techniques, statistical intuition, shells, and more
 - Designed and developed distributed data apps with Scala, Spark, and Python
 - Designed and developed AWS infrastructure with CDK
- May 2019–Dec 2019 **Software Developer**, *Jonah Group*, Toronto, Canada.
Full Time
- Developed Spring REST API for RouteOne LLC
 - Developed Spring REST API and React front end for the R&D team ML Recommender System project
 - Created IoT image recognition deployment software on the R&D team, using Python on a Raspberry Pi
- Sep 2016–Apr 2017 **Teaching Assistant**, *University of Toronto*, Mississauga, Canada.
Jan 2019–Apr 2019
- Computer Organization (CSC258): Taught digital logic labs and facilitated tests
 - Programming on the Web (CSC309): Taught tutorials, helped students with web programming based assignments, and graded assignments
 - Introductory Physics (PHY136): Taught and graded labs, wrote, administered, and graded quizzes
- Mar 2018–Dec 2018 **Software Developer**, *Carpages.ca*, Remote / Mississauga, Canada.
Contract
- Ported existing vehicle search Android app from Java to Kotlin
 - Developed improvements to the vehicle search apps (iOS and Android)
 - Improved the Carpages REST API, with the Symfony 2 PHP Framework
 - Worked on the mobile dealer inventory management system using the JavaScript React Native framework
- May 2017–Feb 2018 **Embedded Software Developer**, *Verity Studios AG*, Zürich, Switzerland.
Internship
- Improved drivers, bootloaders, and system functions for the Verity indoor-positioning systems and flying machines with ARM C and C++
 - Improved build scripts and flashing scripts with Python, BASH, and Tcl
- May 2016–Sep 2016 **Software Developer**, *University of Toronto*, Toronto, Canada.
Contract
- Created a Java port of the Programming Classroom Resource System (PCRS)
 - Refactored Django code base to segregate language specific functionality
 - Developed a Java sandbox for securely generating visualizer information from student code ([elimirks/JavaJail](#) on Github)

- Sep 2012–Sep 2015 **Mobile Software Developer**, *Carpages.ca*, Mississauga.
60% Part Time
- Created used car search application with over 60,000 unique users (Android & iOS)
 - Created vehicle inventory management app for dealers (iOS)
 - Contributed to the design and implementation of a PHP RESTful API

Education & Certifications

- Mar 2025–Sep 2025 **NEXT AI Participant-Founder**.
Dec 2020 **AWS Certified Solutions Architect, Associate**.
Sep 2014–Apr 2019 **HBSc, Physics, Computer Science (Double Major)**, *University of Toronto*, Mississauga, CGPA: 3.97/4.00.

Volunteer Experience

- May 2016–Apr 2017 **President**, *University of Toronto Mississauga Mathematical and Computational Sciences Society*, Mississauga.
 - Planned and hosted coding competitions, hackathons, and workshops
 - Organized hiring and administration of the 2016-2017 MCSS team

Oct 2015–Apr 2016 **Research Assistant**, *UTM Nano-Microscopic Biological and Medical Physics Lab*, Mississauga.
 - Improved the Java 3D ImageJ Viewer for tissue sample imaging
 - Created Matlab scripts for statistical analysis of lab data

Awards

- 2019 **Award for Outstanding Performance in Chemical & Physical Sciences**, *University of Toronto*.
 - Awarded to the top graduating student in Chemical & Physical Sciences.

2016, 2017, 2019 **Dean's List Scholar**, *University of Toronto*.

2016, 2017 **Mathematical & Computational Sciences Honour Roll**, *University of Toronto*.
 - Awarded to students who have obtained a grade of 90% or greater in two or more Mathematics, Computer Science, or Statistics courses at U of T

2015, 2016, 2019 **Chemical and Physical Sciences Honour Roll**, *University of Toronto*.
 - Awarded to students who have obtained a grade of 90% or greater in two or more Chemical or Physical Sciences courses at U of T

2015 **Best Intel Edison Hack**, *WearHacks*.
 - Granted to the team that best utilized the Intel Edison hardware
 - My team built a medication tracking wristband and server to aid physicians in monitoring medication dosages for their patients

Hobby Project Work

- 2023 **Wavy Crockett**.
Wavy Crockett is a software audio synthesizer platform. I wrote a Lisp interpreter in Rust with some fundamental audio building blocks such as sine waves, square waves, white noise, ADSR envelopes, gaussian blurs, etc. Using those building blocks I played around with different audio effects such as LFO envelopes, drum synthesis, chord progressions, and sequencing.
- 2022 **Phlogiston 6502**.
I hand wired a 6502 computer (which I call the Phlogiston 6502) following the Eater 6502 kit. I had a broken Atari 800 I picked up from the scrapyard so I decided to take out the POKEY chip and rewire the Eater 6502 to have POKEY audio synthesis and I/O muxing!
- 2021 **b64: B Compiler Written in Rust**.
I'm interested in computer history and wanted a good Rust project, so I wrote a B compiler (the language that came before C). It only has a single dependency: jemalloc. I wrote the compiler pipeline to be highly parallel, so it runs at 3 million LOC/second. I wrote a recursive descent parser from scratch with some SIMD in the lexing stage "for fun". The code generation + linker emits an ELF x86-64 binary. I also implemented and included a small built in linker and optimizer.